

Intrusion Detection in an Industrial Control System Using Unsupervised Machine Learning Algorithms

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Abstract—In an era characterized by ubiquitous network connectivity, safeguarding the security and integrity of network traffic has become increasingly vital. The MODBUS protocol has become widely used for Industrial Control Systems (ICS) communication. However, it was not originally designed for today's highly connected, internet-exposed environments, thereby leaving it particularly vulnerable to modern cyber-attacks. Therefore, the development of innovative, robust, and efficient intrusion detection systems (IDS) to safeguard data against cyber-attacks has become imperative. This study introduces two unsupervised machine learning algorithms namely, One-Class Support Vector Machines (One-Class SVM) with Principal Component Analysis (PCA) and Isolation Forest (iForest) for intrusion detection. PCA reduces the data's dimensionality while preserving 97.10% of its variance, thus enabling efficient computation with minimal loss of information. To optimize the One-Class SVM, a grid search approach was employed to fine-tune hyper-parameters, specifically the kernel coefficient (γ) and the regularization parameter (ν). A total of 25 parameter combinations were evaluated, resulting in optimal values of $\gamma = 10.0$ and $\nu = 0.07525$, yielding a balanced F1-score of 0.975 on the training dataset. The best-performing model achieved a high recall of 0.976, indicating strong anomaly detection capability, but had a low precision of 0.406, suggesting a high false positive rate. The iForest algorithm on the other hand demonstrated strong anomaly sensitivity, as evident by a recall of 0.985. However, it demonstrated extremely poor precision and overall accuracy thus limiting its practical deployment feasibility.

Keywords— *Anomaly Detection, iForest, One-Class SVM, Grid Search, Hyper-parameter Tuning*

I. INTRODUCTION

Due to the increasing frequency of cyber incidents caused by skilled and persistent attackers, several nations are implementing policies and regulations to ensure adequate security measures are in place to safeguard networks [1], [2]. Historical incidents have demonstrated that breaches in Industrial Control Systems (ICS) can lead to large-scale power outages, the release of hazardous chemicals, and even disastrous explosions. Ensuring the security of ICS is therefore vital for maintaining reliable, safe, and adaptable system performance. In this study, the focus is on the MODBUS industrial communication protocol.

Fundamentally, MODBUS employs a request/response communication paradigm, referred to as a master/slave architecture. Each device participating in the

communication process is assigned a unique address. In this configuration, the master device initiates all communication transactions, referred to as queries, while the slave devices respond by either providing the requested data or executing the specified actions contained within the query [3].

MODBUS/TCP is an adaptation of the MODBUS/RTU protocol, specifically designed for operation over TCP/IP networks. In this variant, standard MODBUS messages are encapsulated within TCP/IP headers and transmitted across Ethernet-based networks, facilitating communication between IP-connected devices. Typically, MODBUS slave devices listen for incoming TCP connections on port 502. Once a connection is established, Protocol Data Units (PDUs) are exchanged within TCP segments, enabling seamless data transfer [4]. The introduction of MODBUS/TCP was driven by the increasing demands of the ICS sector and the need for scalable communication within larger SCADA networks. This protocol supports both communications within a local area network (LAN) based MODBUS environment and across broader IP interconnected networks. Notably, MODBUS/TCP enables multiple concurrent transactions, allowing masters to simultaneously query multiple slave devices, while slaves are capable of handling requests from multiple masters concurrently [5]. It is extensively utilized across a range of industrial applications owing to its real-time responsiveness, high degree of flexibility, and straightforward deployment process [6]. In the TCP/IP implementation, MODBUS operates as application-layer data within the network stack.

Although MODBUS has become the *de facto* standard for ICS; mainly due to its simplicity, open nature, and widespread support from automation vendors, it was never intended to operate in today's highly connected and internet-exposed environments. The MODBUS protocol was originally developed without consideration for cyber-security, as ICS at the time operated in isolated environments where security threats were minimal or non-existent [7]. Consequently, MODBUS lacks essential security mechanisms such as authentication, and authorization, making it highly vulnerable to modern cyber-attacks. Moreover, the protocol's design flaws, and lack of built-in protections have led to numerous exploitable vulnerabilities [5]. In general, attacks targeting MODBUS based systems and networks can be classified into three primary categories. The first includes attacks that exploit inherent vulnerabilities within the MODBUS protocol specifications. The second encompasses attacks that take advantage of flaws or inconsistencies in

vendor-specific implementations of the protocol. Finally, the third category involves attacks directed at the supporting infrastructure, which comprises information technology resources, networking components, and telecommunications systems essential for MODBUS communication [8]. Typical attack vectors targeting industrial communication protocols (ICP) include Man-in-the-Middle (MITM) attacks, Denial-of-Service (DoS) attacks, replay attacks, and various forms of injection attacks, among others [9], [10], [11]. Existing security solutions, such as firewalls and authorization systems, continue to face significant challenges, primarily due to the diversity of protocols and standards they must support, rendering them ineffective against advanced cyber threats [12].

The complex structure of ICS, with their numerous cross-connections and high levels of equipment noise, makes anomaly detection more difficult. Furthermore, the ongoing advancements in ICS require the development of innovative, robust, and efficient strategies to protect data from cyber-attacks. Intrusion Detection Systems (IDSs) constitute an essential layer within cyber-security architectures, offering real-time monitoring of network traffic and system activities to identify potential threats or policy violations. The primary objective of an IDS is to detect malicious behaviors that may compromise the availability, integrity, or confidentiality of information systems; capabilities that extend beyond the scope of conventional firewalls. Functioning as either software or hardware-based solutions, IDSs enhance situational awareness and enable timely responses to security incidents. Broadly, IDSs are classified into two categories: Signature-based Intrusion Detection Systems (SIDS), which detect threats by matching observed activity against a database of known attack patterns; and Anomaly-based Intrusion Detection Systems (AIDS), which identify suspicious behavior by detecting deviations from established norms [13].

Machine learning is increasingly recognized as a practical solution for addressing various business and societal challenges involving data processing. It utilizes algorithms that learn iteratively from input data to build a model of the system and predict outcomes, all without the need for explicit programming [14]. These systems are designed to monitor and detect intrusions or malicious activities within ICS environments, and the security mechanisms can be categorized as supervised, unsupervised, or semi-supervised [15]. The One-Class Support Vector Machine (One-Class SVM), also known as unary classification, is an unsupervised machine learning technique that has gained considerable attention for its effectiveness in anomaly detection [16], [17], [18]. It is a variant of the traditional Support Vector Machine (SVM), which is a supervised machine learning algorithm also utilized for anomaly detection [19]. Yau *et al.* (2017) [18] used the semi-supervised One-Class SVM algorithm to detect anomalies in a simulated traffic light control system, with training data collected from the Programmable Logic Controllers (PLC) system under normal conditions. The experiment showed a training accuracy of 98%, with test results of 78%, 75%, and 82%, leading to an overall recall value of 78%. These findings highlight the potential of One-Class SVM for anomaly detection in a PLC environment, though further research is needed to fully explore its effectiveness.

Hyper-parameter selection is critical in defining the behavior and performance of the One-Class SVM model for anomaly detection [20]. In this study, the standard Gaussian

Radial Basis Function (RBF) kernel was used, and two hyper-parameters required optimization: the regularization coefficient, 'nu' (ν) and the Gaussian kernel width, 'gamma' (γ). The 'nu' parameter governs the balance between maximizing the margin and minimizing the error on the training set, controlling the upper limit of rejected (noise) target data during training [16], [21], while the 'gamma' value dictates how closely the model fits the data points in the feature space, influencing the boundary's shape [21]. Optimizing these parameters using methods like grid search is vital to enhance the model's ability to identify anomalies effectively.

Isolation Forest (iForest), unlike traditional anomaly detection techniques that rely on density estimation or boundary modelling, utilizes a fundamentally different approach based on the concept of isolation. The algorithm functions by constructing multiple isolation trees each created through random selection of features and split values between their observed minimum and maximum ranges. This recursive partitioning process proceeds until each data point is isolated or until a defined tree depth is reached. Because anomalous observations tend to require fewer splits to become isolated, they are effectively distinguished from the bulk of the dataset using this method. One of its major advantages is the minimal need for extensive data pre-processing and its ability to deliver rapid inference, which is particularly beneficial for real-time anomaly detection in ICS environments [22], [23]. Furthermore, it is well-suited for intrusion detection in Modbus-based industrial networks due to its ability to operate in an unsupervised manner, eliminating the need for labelled attack data. Additionally, it is robust to noise commonly found in industrial communications and provides interpretable results through short isolation paths, making it a practical and reliable choice for anomaly detection in ICS environments.

In section II, the materials and methods are presented. Section III discusses the main results and efficacy of the method. Section IV concludes the study.

II. MATERIALS AND METHODS

The proposed methodology establishes a structured framework for anomaly detection in an ICS setting. The approach follows a systematic workflow [24], as illustrated in Fig. 1.

Data was collected from a laboratory test-bed and meticulously pre-processed to maintain its integrity and suitability for anomaly detection. A One-Class SVM model was then fine-tuned through hyper-parameter optimization to enhance detection performance. Comparative analysis was conducted with the iForest algorithm. Finally, after pre-training and testing the models, a post-evaluation phase was conducted to generate results and assess the model's performance.

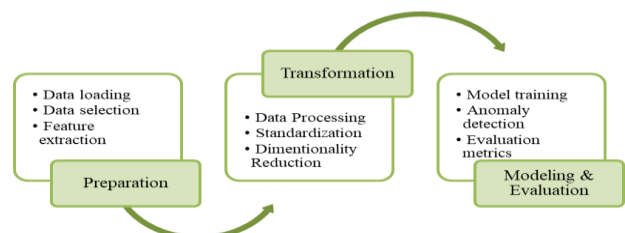


Fig. 1. Workflow for anomaly detection in Industrial Control System (ICS) network traffic data

A. Dataset

The MODBUS TCP SCADA #1 dataset (IEEE DataPort) was used to test the algorithms. The dataset was developed in a process automation environment using MODBUS/TCP equipment and consisted of time-series data. The dataset consisted of the following types of data traffic:

- Clean packets.
- Man-in-the-middle.
- MODBUS Query flooding.
- Ping Flood DDOS.
- TCP SYN Flood DDOS.

For the purposes of this study, ‘Clean packets’ and ‘Man-in-the-middle’ datasets were used. The Clean packets dataset was used to train the algorithm and comprised of 427 842 records (102 Mbytes of data). The test dataset contained both normal and anomalous traffic, comprising of 35 476 records (8.49 Mbytes of data) [24].

B. Pre-processing of data

The data pre-processing phase was crucial for transforming the dataset into a clean and structured format, making it suitable for anomaly detection analysis. The pre-processing process involved six systematic steps, namely:

1. Data loading,
2. Feature selection,
3. Extraction and splitting of MAC and IP addresses,
4. Data conversion and cleaning,
5. Data standardization, and
6. Dimensionality reduction.

Initially, the dataset was examined using the ‘df.info()’ function in the *pandas* library, which provided a concise summary of the data frame and offering insights into its structure, features, and memory usage. The training and testing datasets were successfully loaded, with the training dataset consisting of 427,842 entries, while the testing dataset contained 35,476 entries. This indicated a substantial dataset with a variety of features suitable for modelling. All relevant columns were of object data types, with five columns containing missing values. Specific features were selected for further processing, and functions were created to extract and split the MAC and IP addresses. Each MAC address was divided into six separate octets, while each IP address was split into four octets, generating a total of 10 new features. Alphanumeric characters were handled specifically in the context of MAC addresses. Each octet was converted from hexadecimal to integer. If an octet was missing or malformed, it was replaced with zero imputation. The resulting octets from the MAC address were stored in the ‘mac_matrix,’ and those from the IP address were stored in the ‘ip_matrix.’ A function was applied to horizontally combine the ‘mac_matrix’ and ‘ip_matrix’, resulting in two separate datasets, namely: ‘combined_matrix_train’ and ‘combined_matrix_test’. These combined matrices were then standardized using the ‘StandardScaler’ function to normalize the feature distributions across both datasets. Standardization ensured that each feature contributed equally to the PCA results, mitigating the impact of varying scales. Subsequently, PCA was applied to reduce the dimensionality of the data, preparing it for input into the One-Class SVM and iForest models. This is shown in Fig. 2.

III. RESULTS AND DISCUSSION

A. Pre-processing of data

The mean values of the standardized training data are approximately zero across all features, indicative of successful standardization. The training set had standard deviations close to 1 for significant features, while the majority of features were centered effectively, further confirming that the data was properly scaled. Notably, certain components show 0 standard deviation, typically indicating constant features or features with no variance. The mean standardization of the testing data ranged from 0.0058 to 0.0798, highlighting slight deviations from the training data mean. The standard deviations of the testing data were very close to 1, indicating that the test data maintained a similar scale after standardization.

PCA was applied to reduce the dimensionality, culminating in a transformed dataset termed ‘combined_matrix_train’ which contained the following data shapes: the processed training data contained 427 842 entries and 10 columns, and the processed test data contained 35 476 entries and 10 columns. Table 1 indicates the 10 principal components and the variance explained by each of these components. It can be noted; that the first two principal components account for 97.10% of the total variance in the dataset, suggesting that a significant portion of the original data’s information can be retained using just the first two components for further analysis or visualization.

B. Hyper-parameter Tuning

A grid search was conducted to identify the optimal hyper-parameters for the One-Class SVM model. The parameters ‘gamma’ and ‘nu’, which control the kernel’s influence and the margin of tolerance for anomalies, respectively, were varied across a range of values. Table 2 shows the grid search using ‘gamma’ values ranging from 0.001 to 10 (on a logarithmic scale) and ‘nu’ values from 0.001 to 0.1 (on a linear scale). At low ‘gamma’ values (ranging from 0.001 to 0.01), the models kernel is broader, leading to smoother decision boundaries. The results indicate that as ‘nu’ increases, the precision and recall also improve, with a peak F1-score of 0.831 observed at gamma = 0.001 and nu = 0.100. For intermediate gamma values (ranging from 0.1 to 1.0) the decision boundary displays more sensitivity to the data.

Table 1. Tabulation of the 10 principal components and the explained variance of each component.

Principal Component	Explained Variance (%)
C 1	68.56
PC 2	28.54
PC 3	1.96
PC 4	0.69
PC 5	0.21
PC 6	0.036
PC 7	0.000
PC 8	0.000
PC 9	0.000
PC 10	0.000

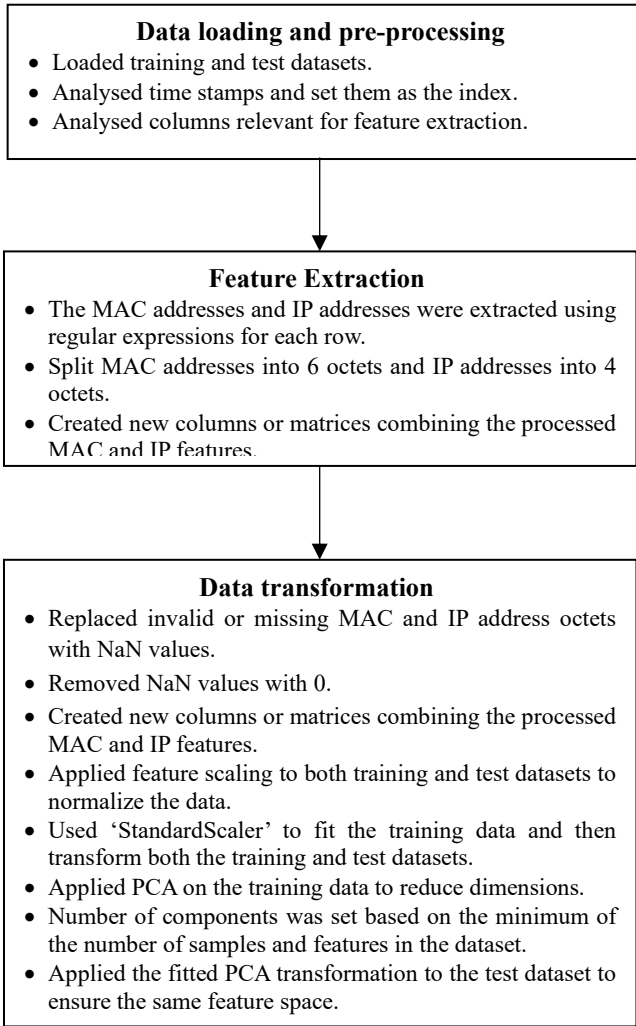


Fig. 2. Data pipeline illustrating the process workflow for data pre-processing

C. Unsupervised Machine Learning Algorithms

Notably, at $\gamma = 1.0$ and $\nu = 0.001$, the One-Class SVM model achieved a high F1-score of 0.845 with a recall of 0.893. This configuration exhibited balanced performance across all metrics. At the highest γ value tested, the model showed significant improvement. The combination of $\gamma = 10.0$ and $\nu = 0.07525$ yielded the best overall results among all configurations that were applied to the training dataset. This combination proved to be the most effective, achieving the highest accuracy of 0.951, precision, recall, and an F1 score of 0.975.

The best-performing model utilizing the testing dataset achieved a recall of 0.976. Thereby indicating that most anomalous instances were successfully identified as indicated by Table 3. However, the evaluation results from the testing data yielded a precision value of 0.405, signifying that less than half of the instances predicted as anomalies were actual outliers. This low precision points to a significant issue with false positives, where normal data is misclassified as anomalous. Conversely, the recall value was notably high at 0.976, indicating that the model is proficient at detecting true anomalies. Such a high recall is essential in anomaly detection scenarios as missing a genuine anomaly could have severe implications for operational integrity.

iForest also resulted in a high recall of 0.98, indicating that the model was highly effective in identifying anomalous instances. However, this comes at the cost of very low

precision of 0.098, thereby suggesting that the model generated many false positives. This trade-off is further reflected in the F1-score of 0.179, which balances precision and recall. Despite the model's strong sensitivity, the low precision significantly reduces its practical implementation. The overall accuracy is 0.113, which is expected in highly imbalanced datasets, but still underscores the model's limited ability to correctly classify the majority class in data normal traffic. The model's high recall value is critically important in the context of anomaly detection, especially within ICSs where the cost of missing a true anomaly can be substantial. A recall this high indicates that the model is effective at identifying the vast majority of actual anomalies, which is crucial for maintaining security and operational integrity. The high recall minimizes the risk of overlooking genuine threats, such as cyber intrusions or system failures, which could lead to severe consequences including data loss, operational downtime, or safety incidents. While having a high recall is essential, it is equally important to balance it with precision to ensure that the system is reliable and accurate.

The F1-score for the One-Class SVM was 0.573, indicating that despite the model's effectiveness in capturing many anomalies, it struggled with high false positives, thereby adversely impacting its overall performance. These findings highlight the importance of balancing precision, recall, and operational needs when selecting a technique. Agyemang *et al.* [23] conducted a similar study using the One-Class SVM algorithm to detect anomalies in a synthetically simulated dataset. The results depicted a similar trend observed in this study, achieving a balanced performance with recall=100.00% and F1 score=66.67%, demonstrating robust performance in accurately identifying true outliers.

However, the moderate precision of 50.00% suggested a tendency to misclassify certain normal data points as anomalies. These findings highlight the importance of balancing precision, recall, and operational needs when selecting a technique. Lee *et al.* (2023) [25] reported on the effect of hyper-parameters on One-Class SVM decision boundaries. Too large 'nu' values and the model tends to exclude too much data from the decision boundary, thus poorly performing in outlier detection. Conversely, they reported a large 'gamma' value distorts the decision boundary from the training data, causing over-fitting.

IV. CONCLUSION

In this study, the best performing model was identified with $\gamma = 10.0$ and $\nu = 0.07525$, which produced an F1-score of 0.975. However, the findings from the test dataset indicated that the model was over-fitting, similar to the results reported by Lee *et al.* (2023) [23]. This research effort highlighted the critical role of hyper-parameter tuning in the performance of the One-Class SVM model for anomaly detection. The study demonstrated that with appropriate preprocessing, feature extraction, and dimensionality reduction, One-Class SVM is an effective tool for anomaly detection in network traffic data. The combination of optimal hyper-parameter selection and robust evaluation metrics provided a comprehensive understanding of the model's strengths and areas for improvement. These results are promising for applications in real-world network security scenarios where the detection of rare and subtle anomalies is critical. Future work may explore additional techniques to enhance its performance and further refine the model's ability to minimize false positives while maintaining high recall.

Table 2. Grid search of different hyper-parameter (Gamma and nu) values and the results of the various metrics.

Hyper-parameters		Precision	Recall	F1-score	Accuracy
gamma (γ)	nu (ν)				
0.001	0.00100	0.177014	0.177479	0.177246	0.699318
0.001	0.02575	0.184101	0.184584	0.184342	0.701911
0.001	0.05050	0.798528	0.809024	0.803742	0.684801
0.001	0.07525	0.675335	0.674968	0.675151	0.562549
0.001	0.10000	0.830879	0.830879	0.830879	0.719021
0.010	0.00100	0.055392	0.045543	0.049987	0.899284
0.010	0.02575	0.186874	0.187365	0.187119	0.702926
0.010	0.05050	0.187182	0.187674	0.187428	0.703039
0.010	0.07525	0.179325	0.179796	0.179560	0.700163
0.010	0.10000	0.215178	0.205209	0.210075	0.667606
0.100	0.00100	0.179501	0.104450	0.132057	0.744334
0.100	0.02575	0.838705	0.757258	0.795903	0.674202
0.100	0.05050	0.382816	0.291636	0.331063	0.548794
0.100	0.07525	0.809649	0.738554	0.772469	0.647762
0.100	0.10000	0.216334	0.129156	0.161746	0.695851
1.000	0.00100	0.801643	0.892941	0.844833	0.737372
1.000	0.02575	0.811144	0.821313	0.816197	0.700586
1.000	0.05050	0.042075	0.109176	0.060742	0.857904
1.000	0.07525	0.202411	0.191943	0.197038	0.683645
1.000	0.10000	0.206968	0.196264	0.201474	0.685393
10.000	0.00100	0.933764	0.947316	0.940491	0.888009
10.000	0.02575	0.066173	0.139173	0.089698	0.820921
10.000	0.05050	0.839985	0.917601	0.877079	0.783995
10.000	0.07525	0.974665	0.974552	0.974608	0.950502
10.000	0.10000	0.224198	0.214029	0.218996	0.652779

Table 3. Comparison of the One Class-SVM and iForest (IF) model performance.

Model	Precision	Recall	F1-score	Accuracy
One Class-SVM	0.405	0.976	0.573	0.410
IF	0.098	0.985	0.179	0.113

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